

Natural Gas Market Structure & Basis Dislocation Analysis
AECO vs Henry Hub | Supply & Trading Framework

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Focus: North American Natural Gas Markets
Date: May 2026

1. Objective

To evaluate how regional fundamentals, storage dynamics, and seasonal demand cycles drive natural gas basis behavior between AECO and Henry Hub, and to structure these relationships into a framework for identifying regime shifts and directional market bias.

This framework is intended as a simplified representation of how physical market conditions translate into pricing behavior in North American gas markets.

2. Core Market Structure View

Natural gas pricing is driven by the interaction of:

- Regional supply-demand imbalance (AECO vs Henry Hub)
- Storage injections and withdrawals
- Seasonal demand cycles (winter heating vs summer injection periods)
- Short-term volatility and liquidity conditions
- Weather-driven demand shifts (HDD / CDD impacts on consumption)
- Regional transport constraints (pipeline capacity / AECO takeaway limitations)
- Power market linkage (gas-to-power price transmission dynamics)

These factors collectively determine ****basis volatility and regional pricing dislocations across North American hubs****.

3. Key Analytical Constructs

A. Basis Spread (AECO – Henry Hub)

- Measures regional pricing dislocation
- Widens during supply constraints, transport limitations, or regional imbalance
- Compresses under well-supplied or balanced conditions

B. Storage Regime

- High storage → bearish pressure on forward curves
- Low storage → tightening supply expectations and bullish pressure
- Used as a proxy for physical market tightness

C. Seasonal Regime

- Winter: structurally bullish demand environment (heating load)
- Summer: injection-driven conditions with potential oversupply pressure

D. Volatility Regime

- Rolling volatility used to identify stress periods and regime shifts
- Volatility clustering often aligns with transitions in physical fundamentals and positioning

4. Key Insight

Periods where storage stress coincides with seasonal transitions tend to produce the most significant basis dislocations between AECO and Henry Hub, as regional constraints and shifting demand expectations amplify pricing divergence from underlying fundamentals.

These conditions are most relevant for:

- Short-term directional positioning
- Relative value / basis trading
- Fuel cost optimization in generation portfolios

5. Framework Output

The model produces:

- Market regime classification (Bullish / Bearish / Neutral)
- Basis dislocation signals
- Volatility event detection
- Directional bias indicator
- Simple backtested signal performance metrics

6. Application to Supply & Trading

This framework is directly relevant to:

- Natural gas procurement strategy
- AECO basis risk monitoring
- Fuel cost optimization for gas-fired generation assets
- Short-term trading and hedging decision support

7. Summary

This framework structures natural gas pricing through a regime-based lens, linking physical market fundamentals to observed pricing behavior across key North American hubs, with a focus on identifying periods of pricing dislocation and shifting market conditions.